



SIMATS SCHOOL OF ENGINEERING

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B.E./ B.Tech DEGREE EXAMINATION
UNIVERSITY PRACTICAL EXAMINATIONS
CSA12 - COMPUTER ARCHITECTURE

REGISTER NO: _____

TOTAL MARKS (100) : _____

SET 1

1. Write an assembly language program for subtraction of two 8-bit data $A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$ and $B_7 B_6 B_5 B_4 B_3 B_2 B_1 B_0$ using 8085 processor for $A=38$ and $B=12$ using GNUSim.
2. Logisim is a tool to design the logic gates for Full adder and implement the Full adder using only NAND gates using simulator
3. Using C, Find the smallest number in the given set of numbers. The set of integers given as input are 18,45,66,23,19,10,52 and 71.
4. The 3 stage pipelining includes Fetch, Decode and Execute. Calculate the total clock cycle required to perform the arithmetic operations. Use any high level language to depict the above said situation and predict the result



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SET 2

1. The user gives the input in the binary format. Write a program in C to convert the given number to X_{16} format. Assume the input given by the user is 1100111_2 .
2. A software has simple toolbar to simulate the circuit design is Logisim. Identify the logic gates for the implementation of full adder of your choice. There are 3 inputs A, B and C with Sum and Carry as the output. The result should be tested with the Full adder Truth table.
3. Given two numbers are 66 and 34. Write an assembly language program using GNUsim, to find the Smallest among these two numbers
4. Pipelining is the process of storing and prioritizing computer instructions that the processor executes. Two stage pipelining includes the steps Fetch and Execute. Using 2 stage pipeline concept, identify the clock cycles needed to perform the addition of two numbers say 8 and 10 using any high level language.



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SET 3

1. The performance of the CPU is basically depends on response time, throughput and execution time of a computer system. Write a program to find the CPU performance of a processor using any high level language
2. Logisim is a tool to design the logic gates for Full adder and implement the Full adder using only NAND gates using simulator
3. Write an assembly language program for adding two 8-bit data A7 A6 A5 A4 A3 A2 A1 A0 and B7 B6 B5 B4 B3 B2 B1 B0 using 8085 processor for A=14 and B= 23 using GNUSim
4. The user gives the input in the binary format. Write a program in C to convert the given number to X_{16} format. Assume the input given by the user is 1100111_2 .

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SET 4

1. The user is giving 11001101_2 as input and they are in a need of its equivalent in Octal format. Write a program to perform this conversion using any high level language
2. Get the decimal input 28_{10} from the user. This is to be converted to Hexa decimal number. Write a program to convert Decimal number to Hexadecimal number using any high level language
3. Logisim is a tool to design the logic gates for Full adder and implement the Full adder using only NAND gates using simulator.
4. The factorial of a non-negative integer n is denoted by $n!$, is the product of all positive integers. Using any high level language, write a program to find the factorial of 7 and display the result.



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SET 5

1. Two numbers are given as 24 and 23. Write an assembly language program using GNUsim to find the Greatest among the given two numbers
2. The pipeline is a "logical pipeline" that lets the processor perform an instruction in multiple steps. Two stage pipelining includes the steps Fetch and Execute. Using 2 stage pipeline concept, identify the clock cycles needed to perform the Subtraction of two numbers say 20 and 15 using C language.
3. Logisim is an educational tool for designing and simulating digital logic circuits. With its simple toolbar interface and simulation of circuits as you build them, it is simple enough to facilitate learning the most basic concepts related to logic circuits. Now, identify the logic gates needed to implement half adder and draw the circuit diagram using Logisim simulator. Finally test your design using the truth table. Using Logisim, Design and implement 3-bit Full adder by designing the circuit diagram only using NAND gates. Check the resultant using the Truth Table
4. Given two numbers are 66 and 34. Write an assembly language program using GNUsim, to find the Smallest among these two numbers



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SET 6

1. Using C, Find the smallest number in the given set of numbers. The set of integers given as input are 18,45,66,23,19,10,52 and 71.
2. Write an assembly language program for subtraction of two 8-bit data A7 A6 A5 A4 A3 A2 A1 A0 and B7 B6 B5 B4 B3 B2 B1 B0 using 8085 processor for A=38 and B=12 using GNUSim
3. The user is giving 110011012 as input and they are in a need of its equivalent in Octal format. Write a program to perform this conversion using any high level language.
4. The pipeline is a "logical pipeline" that lets the processor perform an instruction in multiple steps. Two stage pipelining includes the steps Fetch and Execute. Using 2 stage pipeline concept, identify the clock cycles needed to perform the Subtraction of two numbers say 20 and 15 using C language.



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SET 7

1. A decimal number is given as an input by the user. Write a program to convert the Decimal number to its binary equivalent using any high level language. Consider the input given by the user is 55.
2. Using GNUSim, Write an assembly language program for multiplication of two 8-bit data $A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$ and $B_7 B_6 B_5 B_4 B_3 B_2 B_1 B_0$ using 8085 processor for 14 and 22.
3. Write an assembly language program to swap two 8-bit data using 8085 processor. The data at 2001 is 6 and at 2002 is 8, swap the data and the address should not be changed.
4. Using Logisim, Design and implement 3-bit Full adder by designing the circuit diagram only using NAND gates. Check the resultant using the Truth Table.



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SET 8

1. Two numbers are given as 24 and 23. Write an assembly language program using GNUsim to find the Greatest among the given two numbers.
2. Perform the multiplication of 12 and 4 using 2 stage and 3 stage pipelining. Compare the performance of both stage pipelining by considering its clock cycle.
3. A decimal number is given as an input by the user. Write a program to convert the Decimal number to its binary equivalent using any high level language. Consider the input given by the user is 55.
4. The 4 stage pipelining operation includes Fetch, Decode, Execute and Writeback. Calculate the total clock cycle required to perform the Multiplication of two numbers 20 and 12. Use any high level language, depict the above said situation and predict the result.



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SET 9

1. Write an assembly language program for adding two 16-bit data 45 and 12 using 8086 processor implementing by GNUSim
2. When the processing happens in an overlapped manner or introducing parallel execution is called as Pipelining. ARM devices need pipelining because of RISC and emphasize the compiler complexity. Consider each stage in pipelining is equivalent to 1 cycle, that is n stages = n cycles. Consider here $n=3$ because it performs Fetch, Decode and Execute. Implement the design of 3 stage pipeline and calculate the total clock cycles needed to complete 1 instruction Using any high level language
3. Logisim is an educational tool for designing and simulating digital logic circuits. With its simple toolbar interface and simulation of circuits as you build them, it is simple enough to facilitate learning the most basic concepts related to logic circuits. Now, identify the logic gates needed to implement half adder and draw the circuit diagram using Logisim simulator. Finally test your design using the truth table.
4. Greatest Common Divisor is the largest positive integer that divides each of the given integers. Write an assembly language program to find the GCD of 16 and 12.



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SET 10

1. Using GNUsim, Write an assembly language program for division of two 8-bit data $A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$ and $B_7 B_6 B_5 B_4 B_3 B_2 B_1 B_0$ using 8085 processor for 16 by 4.
2. A software has simple toolbar to simulate the circuit design is Logisim. Identify the logic gates for the implementation of full adder of your choice. There are 3 inputs A, B and C with Sum and Carry as the output. The result should be tested with the Full adder Truth table.
3. When the processing happens in an overlapped manner or introducing parallel execution is called as Pipelining. ARM devices need pipelining because of RISC and emphasizes the compiler complexity. Consider each stage in pipelining is equivalent to 1 cycle, that is $n \text{ stages} = n \text{ cycles}$. Consider here $n=3$ because it performs Fetch, Decode and Execute. Implement the design of 3 stage pipeline and calculate the total clock cycles needed to complete 1 instruction Using any high level language.
4. There are 24 hits and 5 misses in accessing cache, Write a program to find the Hit ratio for the given number of Hits and Misses in Cache memory using any high level language.